

AURA MEETING REPORT

Evolution from V4 to V18 · Dataset Transformation · Model Stack · Deployment Readiness

AURA V17/V18 Enterprise Real Estate Intelligence Operating System

Key metric	Value
Dataset size	1,252,804 rows × 174 columns
Years covered	2019–2025
Municipalities	41
Best deep model test result	$R^2=0.9682$ $RMSE=100.47$ $MAE=77.47$ $MAPE=4.05\%$
Bias-corrected production model	$R^2=0.8308$ $RMSE=231.83$ $MAE=193.56$ $MAPE=9.85\%$
Fair value engine test result	$R^2=0.8703$ $RMSE=203.01$ $MAE=164.21$ $MAPE=8.32\%$
Enterprise checkpoint size	1.28 GB

1. Executive Summary

AURA moved from a small property valuation prototype into a full seller-first enterprise intelligence operating system. The platform now combines deep prediction, municipality intelligence, bias correction, fair-value governance, liquidity modeling, scenario simulation, explainability, deployment infrastructure, and continuous learning.

For the meeting, the most important message is that the project is no longer only about estimating price. It now supports strategic decisions: whether to sell, when to sell, how fast the asset is likely to move, and how the market may react under stress.

- V4 was a basic predictive engine with limited dataset scale and limited governance.
- The current platform works on 1.25M+ rows, 174 features, 41 municipalities, and 2019–2025 coverage.
- The current stack is hybrid: FT-Transformer + LightGBM + XGBoost + CatBoost + meta learning + bias correction.
- The system now produces fair value, confidence, sale probability, TOM, scenario sensitivity, and decision support.
- The notebook outputs prove the system is production-oriented, with checkpointing, telemetry, deployment manifests, and MLOps lifecycle support.

1.1) Dataset Growth Chart

Add a full slide/page on data growth.

The message should be: AURA did not just get a better model; it got a much bigger and more realistic intelligence base.

Use this text:

AURA's dataset evolved from an early V4-scale prototype of roughly 137K rows into a 1,252,804-row enterprise dataset with 174 engineered columns, 41 municipalities, and full 2019–2025 coverage. This growth is not only quantitative; it changed the intelligence regime entirely. The larger dataset allowed municipality-level reasoning, temporal validation, multi-source fusion, and more stable production behavior. The result is a platform that learns regional market structure rather than memorizing a small sample of listings.

What the chart should show:

- V4 vs V17/V18 row count
- 137K vs 1.25M+
- 9x growth approximately
- annotations for municipalities, years, and feature count

Suggested caption:

Figure: Dataset evolution from V4 prototype scale to enterprise-scale real estate intelligence.

2. AURA Evolution: from V4 to V18

Version	Main evolution	Core capability	Why it matters now
AURA V4	Initial Portugal predictive engine	Basic valuation ML pipeline	Prototype stage; limited scale and limited market realism
AURA V5	Ensemble + governance foundations	Multi-model prediction + leakage control	Introduced structural safety and multi-model thinking
AURA V6	Institutional valuation intelligence	Fair-value logic + explainability	Improved trust, governance, and market realism
AURA V7	Production-oriented AI infrastructure	Temporal validation + decision engine	Made the system more production-safe and defensible
AURA V8	Latent enterprise intelligence	FT-Transformer + latent ecosystem	Added deep latent representation learning and hybrid intelligence
AURA V9–V12	Market & macro intelligence	Liquidity + scenarios + strategic engine	Expanded from valuation to execution and stress testing
AURA V13–V15	Deployment + MLOps + governance	Drift, telemetry, retraining, lifecycle	Moved from notebook AI to deployable infrastructure
AURA V18	Unified operating system	Full enterprise orchestration	The final architecture is enterprise-oriented and backend-ready

The evolution is important because it shows that each version solved a specific weakness of the previous one: leakage, instability, weak explainability, weak geography, weak deployment, and weak lifecycle management.

2) Model Comparison Table

Add a compact table with the core models and the final performance numbers.

This is one of the strongest technical proof slides because it shows the system is hybrid, not a single-model notebook. The notebook results show the deep model, boosting model, hybrid blend, latent meta layer, bias correction, and fair value engine all as distinct layers.

Use a table like this:

Layer	Role	Test Performance	Meaning
FT-Transformer	Deep tabular intelligence	$R^2=0.9682$, RMSE=100.47, MAE=77.47, MAPE=4.05%	Best raw predictive engine
LightGBM	Asymmetric boosting	$R^2=0.9567$, RMSE=117.29, MAE=89.86, MAPE=4.68%	Strong structured-market baseline
Hybrid Ensemble	Deep + boosting blend	$R^2=0.9670$, RMSE=102.35, MAE=78.56, MAPE=4.12%	Robust operational compromise
Bias-Corrected Model	Residual correction	$R^2=0.8308$, RMSE=231.83, MAE=193.56, MAPE=9.85%	Improves calibration and realism
Fair Value Engine	Governed valuation	$R^2=0.8703$, RMSE=203.01, MAE=164.21, MAPE=8.32%	Final seller-facing intelligence layer

Suggested caption:

Figure: Hybrid model stack and performance comparison.

What to say:

- the best raw model is the FT-Transformer
- the hybrid blend is the most balanced production path
- the fair-value engine is the output that matters commercially
- the latent meta layer is interesting architecturally but weaker than the best operational path

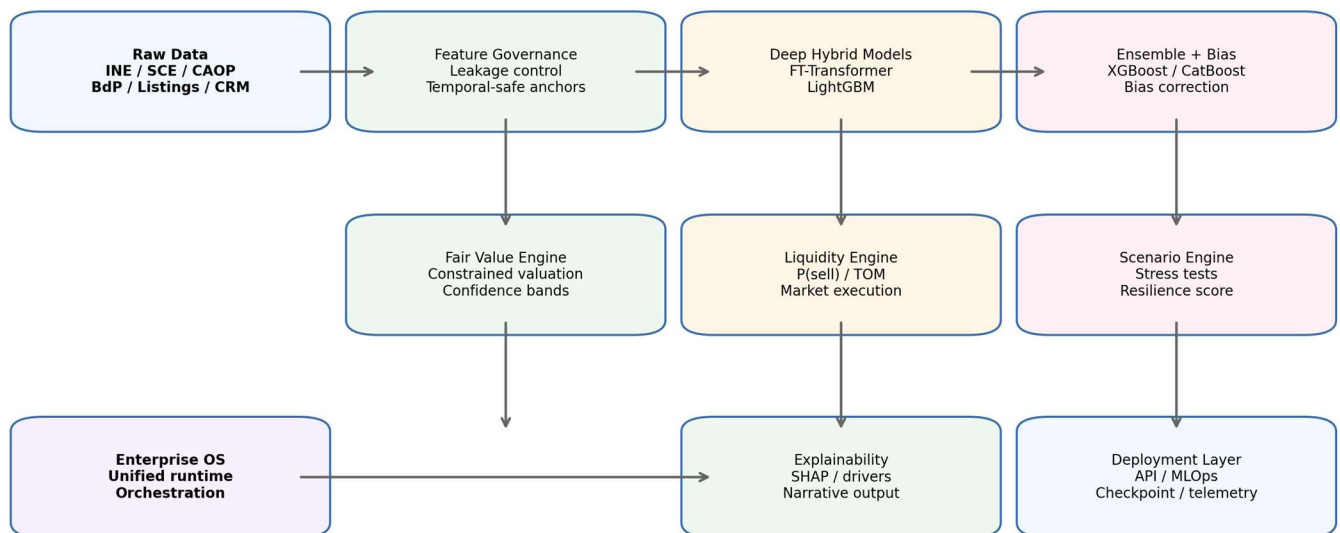
3. The Biggest Change: V4 vs Current AURA

Dimension	AURA V4	Current AURA V17/V18
Dataset size	~137K rows	1,252,804 rows
Feature richness	Limited features	174 engineered features
Geography	District-level and coarse	Municipality intelligence + grid-level reasoning
Time coverage	Narrow historical view	2019–2025 temporal coverage
Modeling	Single predictive approach	FT-Transformer + LightGBM + XGBoost + CatBoost + meta learning

Governance	Minimal or absent	Leakage removal, temporal-safe anchors, bias correction, drift monitoring
Outputs	Price prediction only	Fair value, confidence, sale probability, TOM, scenario impact, decision support
Deployment	Notebook prototype	Enterprise runtime, checkpointing, API gateway, MLOps

This is the strongest comparison slide for the meeting: it demonstrates that the project did not simply improve incrementally — it completely changed category.

AURA Enterprise Architecture — From Data to Operating System



4. Dataset Evolution and Data Categories

The dataset is one of the main reasons the current AURA is much stronger than V4. The system moved from a smaller, district-oriented dataset to a large multi-source real estate intelligence base with spatial, temporal, macroeconomic, and behavioral coverage.

Dataset property	V4 / early stage	Current system
Rows	~137K	1,252,804
Columns	Narrower, simpler feature set	174 columns
Geographic granularity	District-level	Municipality + grid (250m)
Temporal coverage	Limited	2019–2025
Municipalities	Limited	41

Source fusion	Basic	INE + SCE + CAOP + BdP + Listings + CRM
Coverage / integrity	Prototype quality	0 missing target, 0 duplicates, 0 negative prices

The current dataset also gives AURA a market structure that is closer to production reality: no target gaps, no duplicate primary-key rows, and stable temporal coverage across seven years.

Category	What it means	Why it matters
CAT1 property	Basic property attributes	Foundational description of the asset
CAT1.2 calibration	Normalization and calibration signals	Helps align numerical scales and stabilizes learning
CAT2 market	Market dynamics and HPI signals	Captures market momentum and pricing movement
CAT3 macro	Interest rates and macro conditions	Adds economic context and stress sensitivity
CAT4 socio	Demographics and social indicators	Captures demand, density, and population effects
CAT5 geo	Spatial and municipality intelligence	This is the moat; it explains location-driven value
CAT6 behavioral	Buyer/seller and behavioral signals	Helps model decision behavior and market response
CAT7 transaction	Transaction truth and sale history	Ground-truths pricing and market execution
CAT8 POI	Points of interest and accessibility	Adds livability and micro-location quality

CAT5 geo is especially important because the notebook results show municipality features dominating SHAP. That means the platform is learning the local market structure rather than just memorizing generic price patterns.

5. Dataset Growth Visual

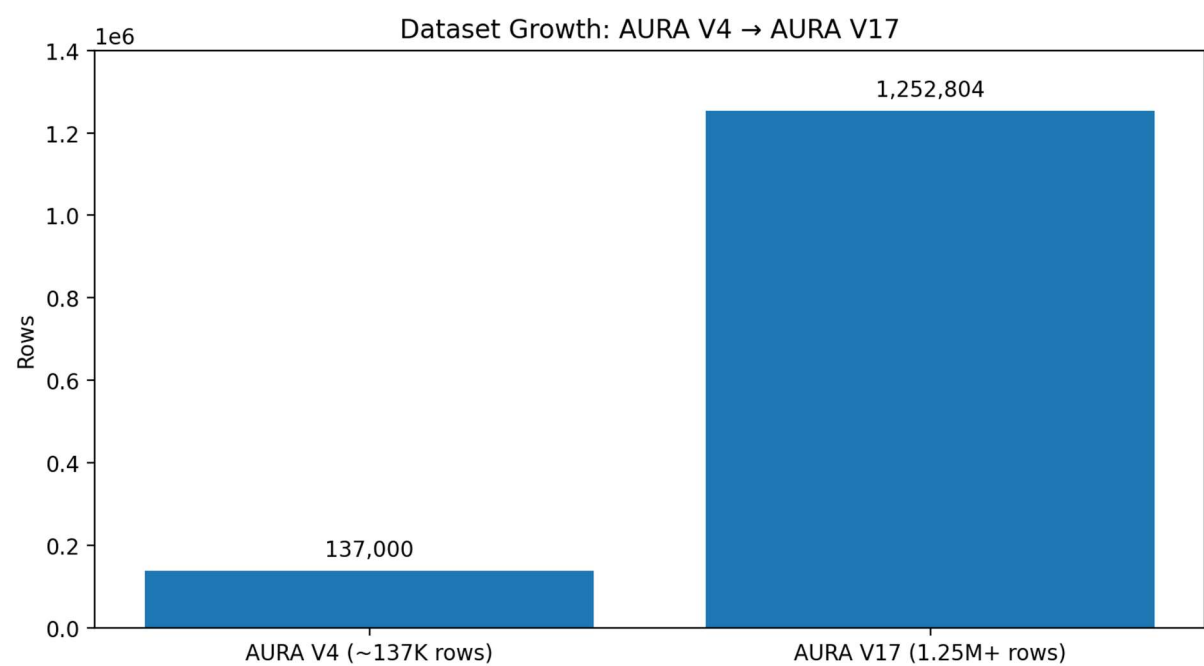


Figure 1. Dataset growth from V4 (~137K rows) to V17 (1,252,804 rows).

The growth is roughly 9.1×. This is not just bigger data — it is a different intelligence regime. The larger dataset enables stronger temporal validation, better municipality coverage, better spatial generalization, and more stable enterprise deployment.

6. Model Stack and Notebook Results

The notebook shows a mature hybrid architecture. The best-performing production path is not a single model, but a combination of deep learning, tree-based models, bias correction, fair-value logic, and governance.

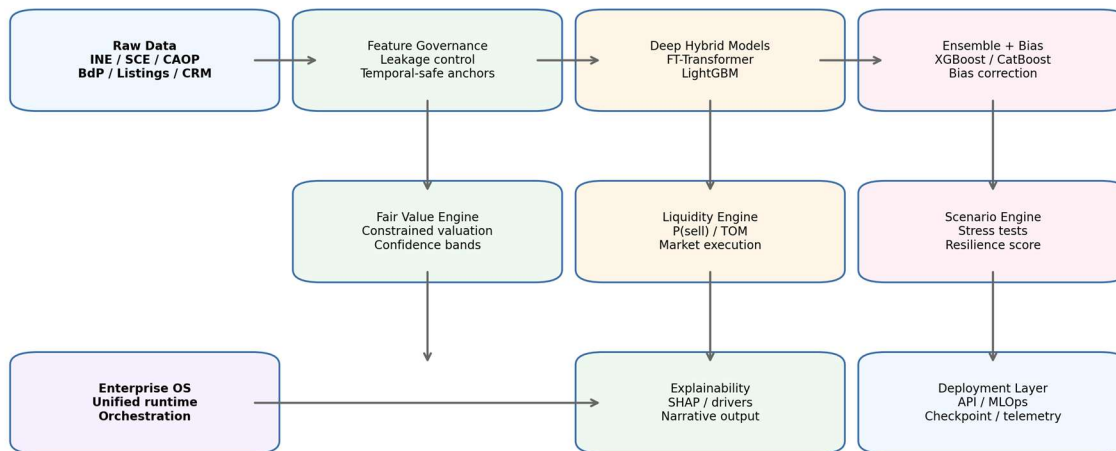
Component	Role	Key notebook result	Interpretation
FT-Transformer	Deep latent model for tabular interactions	Test R²=0.9682 RMSE=100.47 MAE=77.47 MAPE=4.05%	Best raw predictor; captures complex nonlinear interactions
LightGBM	Asymmetric boosting model	Test R²=0.9567 RMSE=117.29 MAE=89.86 MAPE=4.68%	Strong tree model with overpricing penalty
Hybrid ensemble	50/50 deep + boosting blend	Test R²=0.9670 RMSE=102.35 MAE=78.56 MAPE=4.12%	Robust enterprise compromise
XGBoost	Supplementary boosting model	Saved successfully	Adds ensemble diversity
CatBoost	Categorical-aware model	Saved successfully	Useful for categorical structure

Latent meta learner	Meta fusion on latent embeddings	Test $R^2=0.7738$ RMSE=268.09 MAE=228.30 MAPE=11.69%	Exploratory; weaker than the best hybrid path
Bias correction engine	Residual adjustment layer	Test $R^2=0.8308$ RMSE=231.83 MAE=193.56 MAPE=9.85%	Improves realism and calibration
Fair value engine	Constrained valuation + confidence	Test $R^2=0.8703$ RMSE=203.01 MAE=164.21 MAPE=8.32%	Final valuation layer for seller decisions

The best production story is: deep model + boosting + bias correction + fair value governance. The latent meta layer is important as an architectural experiment, but the notebook clearly shows that the strongest operational path is the corrected fair-value stack.

AURA Model Stack — Performance Comparison

Layer	R^2	RMSE	MAE	MAPE	Role
FT-Transformer	0.9682	100.47	77.47	4.05%	Best raw predictor
LightGBM	0.9567	117.29	89.86	4.68%	Strong boosted baseline
Hybrid Ensemble	0.9670	102.35	78.56	4.12%	Most balanced operational path
Bias-Corrected	0.8308	231.83	193.56	9.85%	Calibration + realism
Fair Value Engine	0.8703	203.01	164.21	8.32%	Seller-facing output



Notebook checkpoint / phase	Important result
Rolling temporal validation	$\mu R^2=0.9401 \pm 0.0182$ $\mu \text{MAPE}=5.35\%$
Spatial LOCO validation	$\mu \text{Spatial } R^2=0.7621$
Price segment behavior	Low mean=1378 Mid mean=1817 High mean=2592
Overpricing bias	47.92%
Train-vs-test drift	Mean shift = 419.96 €/m ²
Scenario engine	Resilience score = 2.79/10
Checkpointing	1.28 GB enterprise runtime checkpoint
Runtime readiness	Enterprise runtime validated, deployment artifacts exported

These numbers are meeting-relevant because they show both maturity and honesty: strong predictive performance, visible bias-correction gains, and clear areas for future expansion such as sparse-regime coverage.

7. Visual Evidence from the Notebook

The notebook produced the following visuals, which are the strongest ones to show in the meeting because they explain the system visually in a few seconds.

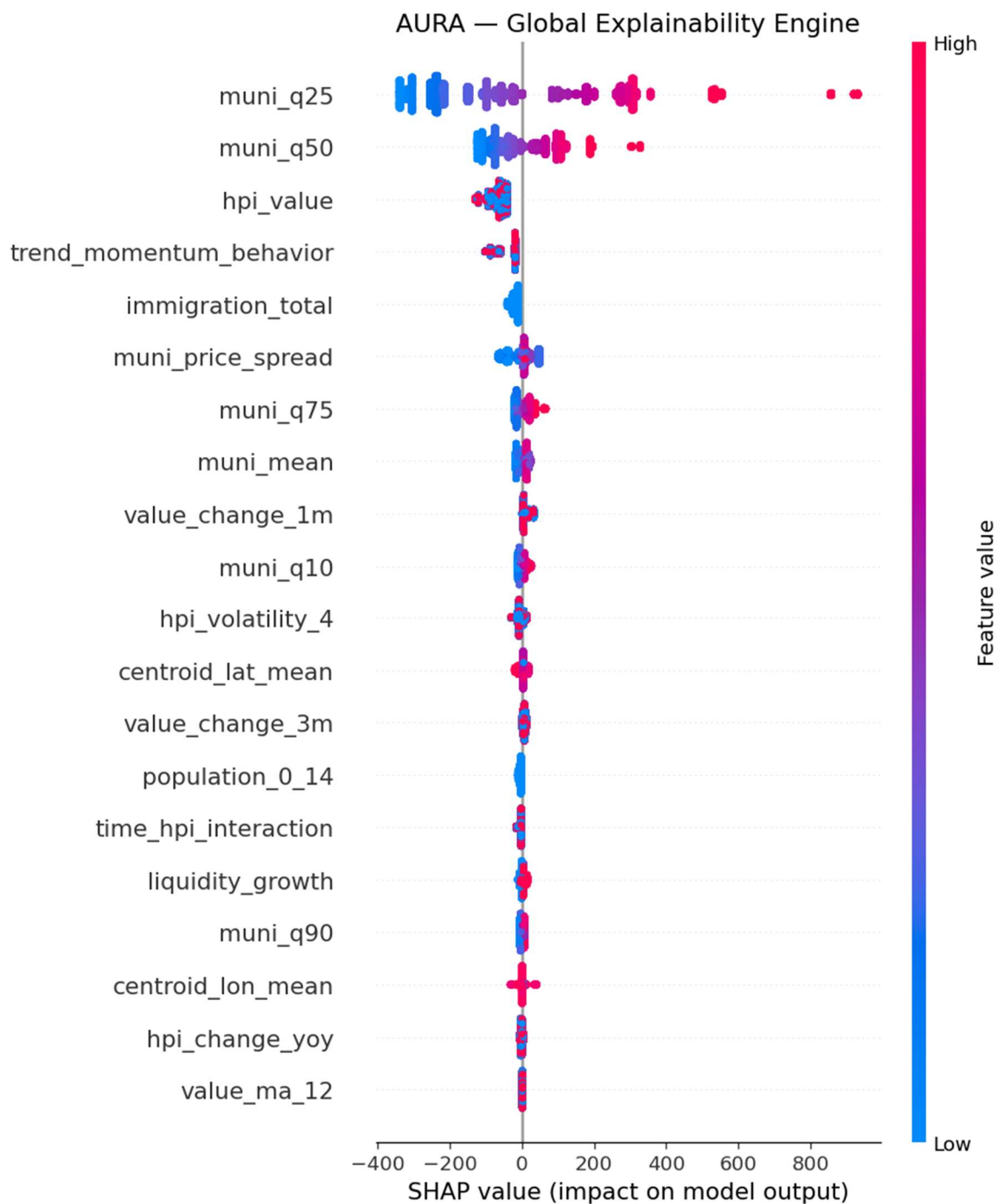


Figure 2. AURA Global Explainability Engine (SHAP). Municipality and market features dominate the explanation.

This SHAP plot is important because it proves that the model is not behaving like a black box. The strongest drivers are municipality quantiles (muni_q25, muni_q50), HPI value, market momentum, immigration, and price spread. That is exactly the story you want in the meeting: location and market structure matter.

SHAP Figure and Explainability Section

This absolutely should be a dedicated section.

The SHAP plot is one of your strongest trust-building visuals because it proves the model is not a black box. In the notebook, the dominant drivers are municipality quantiles and regional market signals, which is exactly the story you want: AURA learns local market structure, not just abstract price formulas.

Use this text:

The SHAP analysis shows that municipality-level quantiles and local market indicators dominate the model's reasoning. This is important because it proves that AURA is learning spatial and municipal market structure, not just generic global price behavior. The strongest drivers include municipality quantiles, HPI value, price spread, market momentum, liquidity growth, and trend behavior. That means the model's logic is aligned with real estate economics: location, market pressure, and temporal dynamics are the main value drivers.

What the figure should communicate:

- municipality importance
- market momentum
- HPI effects
- liquidity effects
- spatial pricing logic

Suggested caption:

Figure: AURA SHAP explainability showing municipality and market structure as the main drivers of value.

Add one sentence explicitly:

- CAT5 geo intelligence is the moat
- the SHAP plot validates that geography really matters

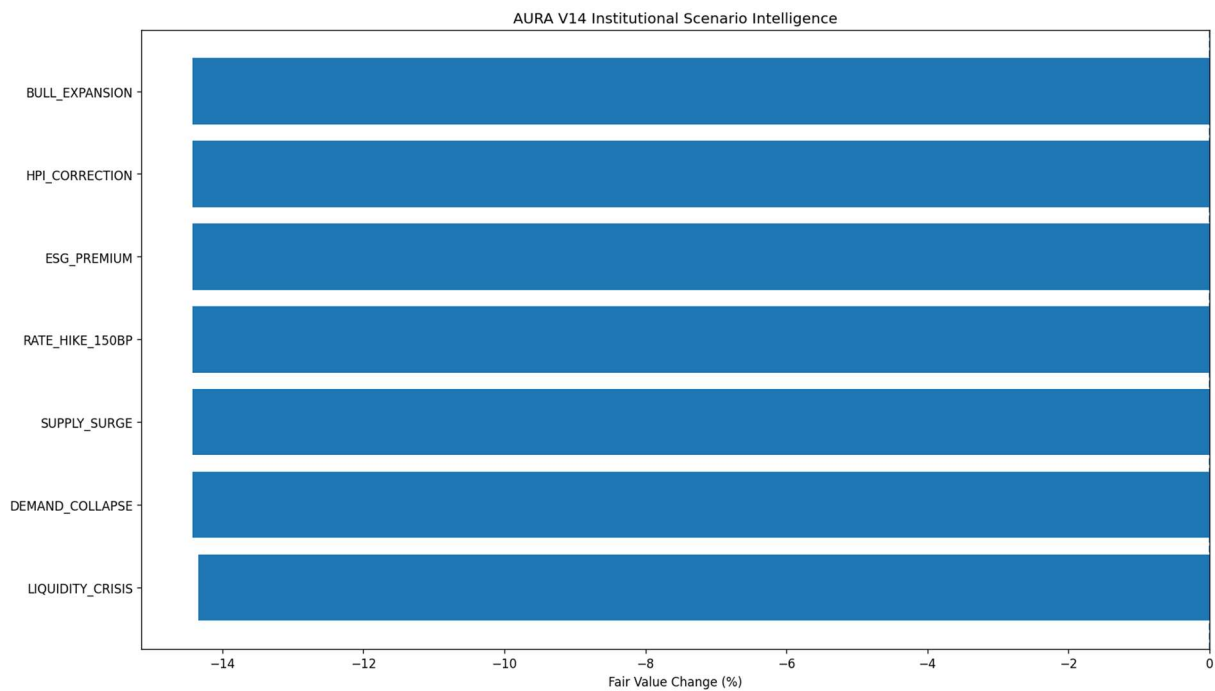
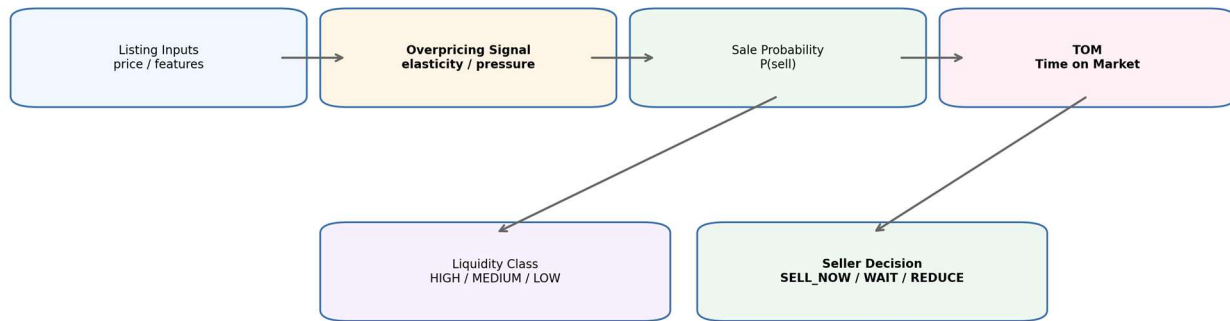


Figure 3. AURA V14 Institutional Scenario Intelligence. Macro shocks stress the valuation engine.

The scenario chart shows how the system reacts to stress paths such as rate hikes, supply surges, demand collapse, liquidity crises, HPI corrections, ESG premium, and bull expansion. The reported resilience score is 2.79/10, which means the system is intentionally sensitive to market shocks and can be used for stress-aware strategic reasoning.

The notebook also includes production runtime and checkpoint outputs showing model export, checkpointing, and enterprise readiness. Those outputs support the story that AURA is no longer just a notebook, but a recoverable runtime.



AURA does not only estimate fair value; it also estimates market execution behavior. The TOM and sale-probability layers answer the seller's real questions: how long the asset may stay on the market, how likely it is to sell, and how overpricing affects execution speed. This turns the platform into seller intelligence rather than only valuation AI. The liquidity layer is what makes the system strategic, because it links pricing to actual market movement and sale feasibility.

9. Current Strengths and Honest Limitations

- Strengths: strong data scale, strong governance, deep model performance, explainability, scenario intelligence, MLOps, and a deployable runtime architecture.
- Strengths: municipality-level features dominate SHAP, proving that the system is learning real location effects rather than generic shortcuts.
- Strengths: the deep model and hybrid ensemble are production-worthy candidates, and the fair-value engine provides a usable seller-facing output.
- Limitation: coverage in the current run is strong for Lisbon, Porto, and Algarve, but interior/islands regime coverage is still sparse in the loaded sample.
- Limitation: the latent meta layer is an interesting architecture layer, but it is not yet the strongest operational path compared with the corrected hybrid stack.

- Limitation: the current work is backend-ready first; the polished frontend/product layer is the next major commercial step.

These limitations are not failures; they are the roadmap. They show exactly where the next value should be created.

10. Final Conclusion

AURA is now much more than a notebook. It is a real estate intelligence operating system with enterprise-scale data, hybrid modeling, explainability, liquidity intelligence, scenario simulation, governance, deployment artifacts, and continuous learning.

For the meeting, the key story is simple: V4 was a prediction prototype; the current AURA is a deployable enterprise intelligence platform. The data grew from roughly 137K rows to 1.25M+ rows, the model stack became hybrid and governable, and the platform now has the architecture needed for backend integration and productization.

Future Work

- Add more deep models and stronger ensembles.
- Enrich features with more spatial, market, and macro signals.
- Fix the missing 7 municipalities.
- Improve municipality-level intelligence.
- Add more scenario and stress tests.
- Strengthen explainability and decision support.
- Validate with the supervisor, then expand the platform.

Appendix: Quick Reference Metrics

Item	Value
Rows	1,252,804
Columns	174
Municipalities	41
Years	2019–2025
Best deep model test R^2	0.9682
Best deep model test RMSE	100.47
Best deep model test MAE	77.47
Best deep model test MAPE	4.05%
Bias-corrected test R^2	0.8308
Fair value engine test R^2	0.8703

Scenario resilience score	2.79/10
Enterprise checkpoint	1.28 GB